1 Input

one fundus photo of the training fold
fundus stage: circle crop, letterbox, CLAHE
▸ **random resized crop to 392 px, scale 0.85–1**

2 Per-photo augmentation, in order

horizontal flip, p 0.5
vertical flip, p 0.5
▸ **rotation ±180°**
colour jitter: brightness 0.7–1.3, contrast 0.7–1.3, saturation 0.8–1.2
colour jitter: saturation 0.7–1.3, hue ±0.05
▸ **gamma U(0.7, 1.4), p 0.5**
greyscale, p 0.1
resolution jitter down to 0.33, p 0.5
affine: translate 0.1, scale 0.88–1.12, shear ±8°
perspective 0.2, p 0.3
Gaussian blur 3×3, σ 0.1–2
▸ **pixel noise σ 0.02, p 0.3**
normalise (imagenet statistics)
random erasing 0.02–0.15 of the area, p 0.35

3 Per batch of 16

▸ **Fourier amplitude swap, centred 3×3 block, random batch partner, p 0.5**

4 Backbone

DINOv2 ViT-S/14 with registers, ImageNet-pretrained (timm vit_small_patch14_reg4_dinov2.lvd142m)
28×28 patches of 14 px; stochastic depth (drop path) 0.1

5 Classification head

dropout 0.3, linear layer to 6 classes

Loss

cross-entropy, class-balanced weights (β 0.999)
▸ **label smoothing 0.1 (B: 0.05)**

Optimiser

AdamW, peak lr 0.0001, layer decay 0.8
weight decay 0.05, gradient clip 1.0
OneCycle: warm-up over 10% of steps, then cosine
20 epochs, batch 16, mixed precision

Head warm-up

▸ **epoch 1: head only, everything else frozen**
OneCycle schedule unchanged

Weight EMA

▸ **decay 0.999 per optimiser step**
▸ **the EMA weights are validated and saved**
last epoch kept, no early stopping

training loop for 4 and 5

▸ bold, coloured outline: differs from arm B (arm H trains with B's recipe)